

Q

Amazon

Given a matrix

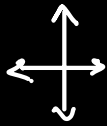
ms

Adabo

0 → empty cell

1 → fresh orange

2 → rotten orange



1	1	0	1	0
0	1	1	2	1
1	2	1	0	1
0	0	1	0	0

day-1 →

1	1	0	1	0
0	1	1	2	1
1	2	1	0	1
0	0	1	0	0

1	1	0	2	0
0	2	2	2	2
2	2	2	0	1
0	0	1	0	0

day-2 ←

1	1	0	2	0
0	2	2	2	2
2	2	2	0	1
0	0	1	0	0

1	2	0	2	0
0	2	2	2	2
2	2	2	0	2
0	0	2	0	0

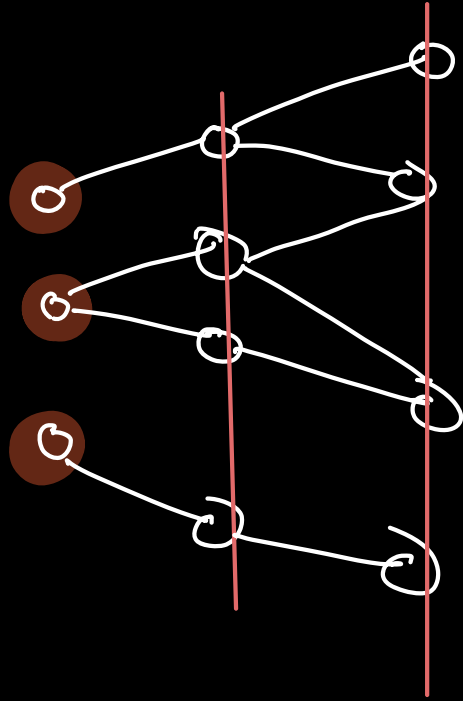
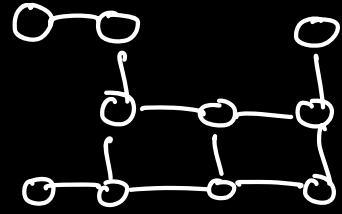
day-3 →

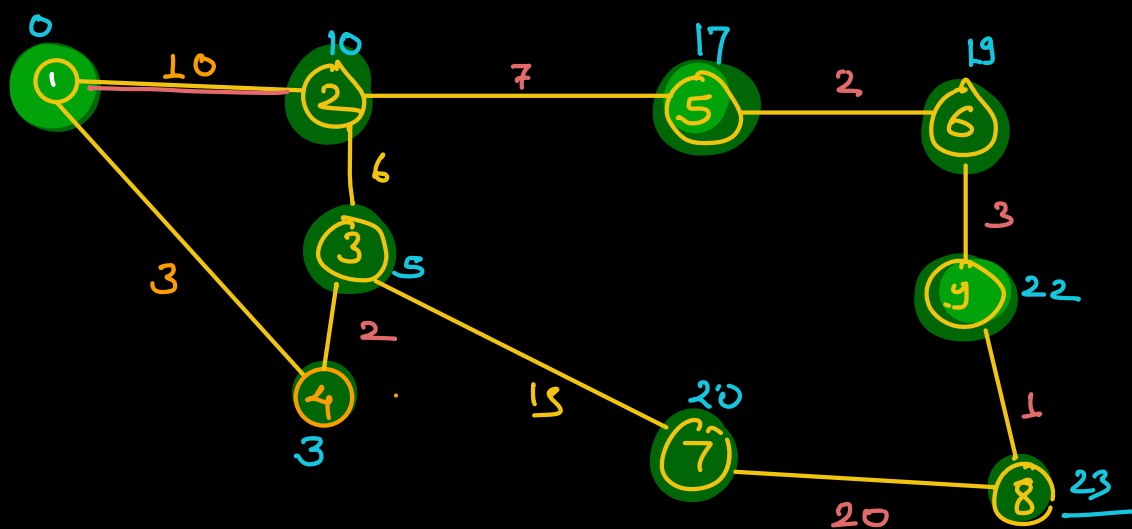
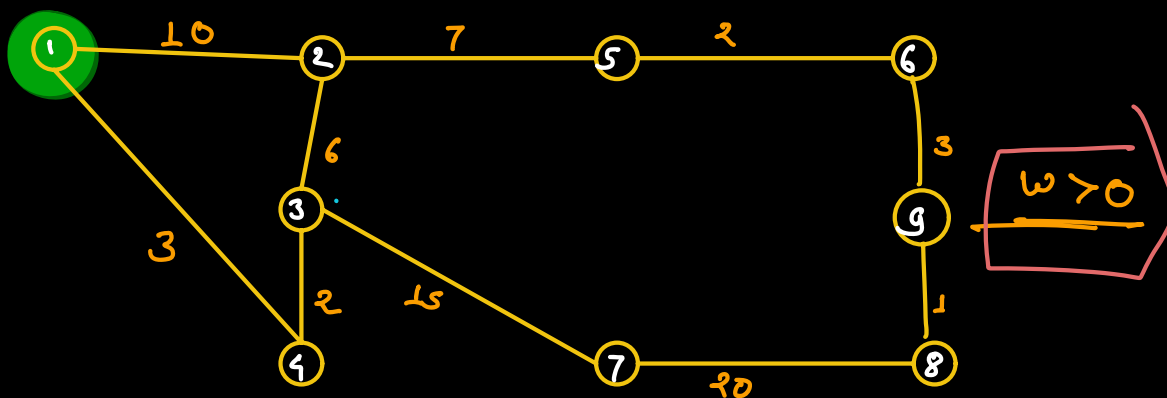
1	2	0	2	0
0	2	2	2	2
2	2	2	0	2
0	0	2	0	0

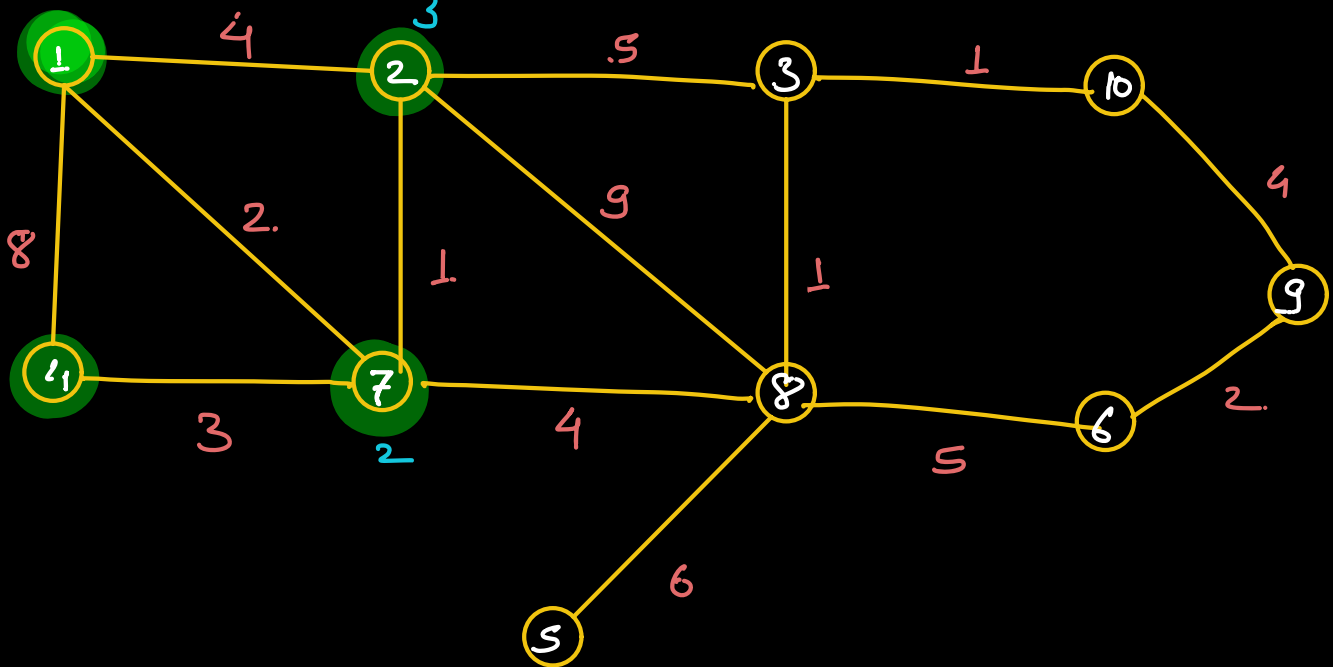
2	0	0	2
0	1	1	0
0	1	1	0
0	0	0	0

10

1	1	0	1	0
0	1	1	2.	1
1	2.	1	0	1
0	0	1	0	0



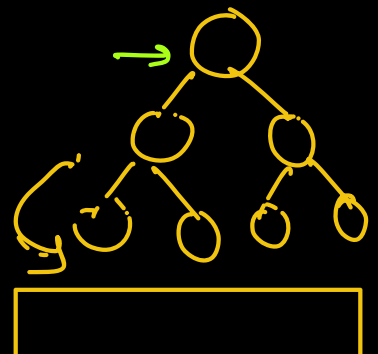
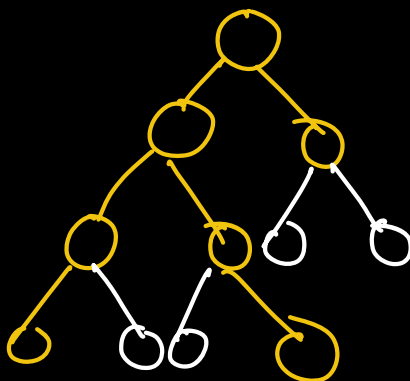




1	2	3	4	5	6	7	8	9
0	∞	∞	∞	∞	∞	∞	∞	∞
	3		5			2		

~~{1, 0}~~ ~~{2, 4}~~ {4, 8} ~~{7, 2}~~ ~~{2, 3}~~ ~~{4, 5}~~ {8, 6}
 {3, 8} {8, 12}

→ PQ



```

PQ < NodeInfo > pq;
dist[N+1]  $\rightarrow \infty$ ;
dist[S] = 0;

```

```

class NodeInfo {
    int node;
    int dist;
}

```

```

pq.add(new NodeInfo(S, 0));
while (pq.size() > 0) {

```

```

    cur = pq.poll(); O(log E)
    u = cur.node;
    d = cur.dist;

```

```

    for each neigh v of u:

```

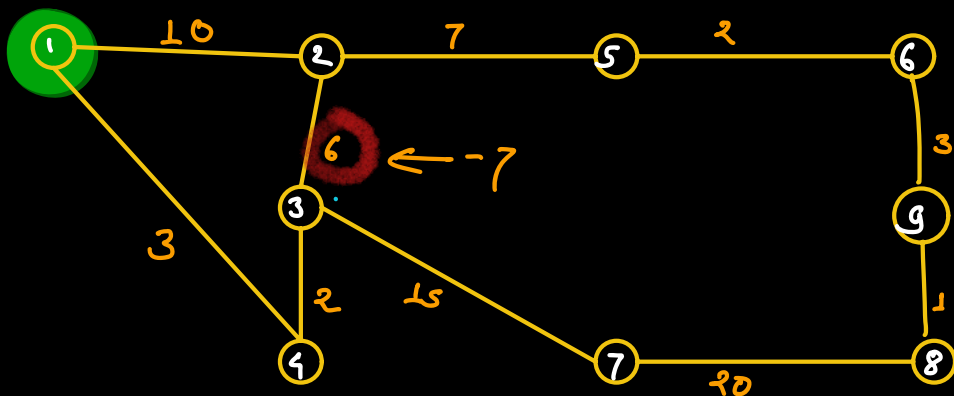
```

        curDist = d + getDist(u, v);
        if (curDist < dist[v]) {
            dist[v] = curDist;
            pq.add(new NodeInfo(v, curDist));
        }

```

$\hookrightarrow O(\log E)$

TC: $\rightarrow$?
SC: ?





$$3, 5 \rightarrow 10$$

$$1, 7 \rightarrow 7$$

;

A	B	C	D
0	3	8	1
	3	8	1

I

$$C - D \quad \infty + 7 \rightarrow \infty$$

$$B - C \quad \infty + 5 \rightarrow \infty$$

$$A - B \quad 0 + 3 \rightarrow 3$$

II

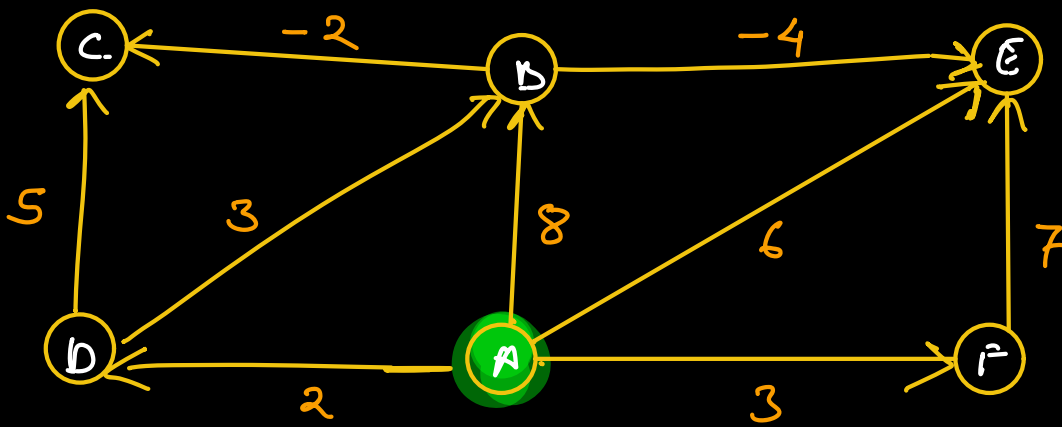
$$\infty + 7 \rightarrow \infty$$

$$3 + 5 \rightarrow 8 \checkmark$$

$$0 + 3 \rightarrow 3$$

III

$$8 + 7 \rightarrow 15$$



A	B	C	D	E	F
0	5	3	2	1	3

A $\xrightarrow{8}$ B
 A $\xrightarrow{2}$ D
 B $\xrightarrow{-4}$ E
 F $\xrightarrow{7}$ E
 A $\xrightarrow{3}$ F
 D $\xrightarrow{5}$ C
 A $\xrightarrow{6}$ E
 B $\xrightarrow{-2}$ C
 D $\xrightarrow{3}$ B

I
 0 + 8 $\rightarrow$ 8
 0 + 2 $\rightarrow$ 2
 8 + (-4) $\rightarrow$ 4
 2 + 7 $\rightarrow$ 9
 0 + 3 $\rightarrow$ 3
 2 + 5 $\rightarrow$ 7
 0 + 6 $\rightarrow$ 6
 8 + (-2) $\rightarrow$ 6
 2 + 3 $\rightarrow$ 5

II
~~5~~
~~3~~
 5 + (-4) $\rightarrow$ 1 $\leftarrow$
 3 + 7 $\rightarrow$ 10 X
 0 + 3 $\rightarrow$ 3
 2 + 5 $\rightarrow$ 7
 0 + 6 $\rightarrow$ 6 X
 5 + (-2) $\rightarrow$ 3
 2 + 3 $\rightarrow$ 5

III
 X
 X
 X
 X
 X
 X
 X
 X

